

Tsurugi Digital Forensics – Browser Artifact Analysis

Introduction

The goal of this lab is to gain hands-on experience with digital forensics by using Tsurugi Linux to examine browser artifacts and determine what can be learned from a user's browsing history. Tsurugi was selected because it is a Linux distribution designed specifically for digital forensics and incident response (DFIR) and comes equipped with a large collection of forensic analysis tools.

For this investigation, I chose Hindsight, a tool designed to analyze Chrome and Chromium browser artifacts. Using a sample Chrome history database, I will use Hindsight to extract and examine browsing activity, establish a timeline, identify searches and visited services, and determine what conclusions can reasonably be supported by the available evidence.

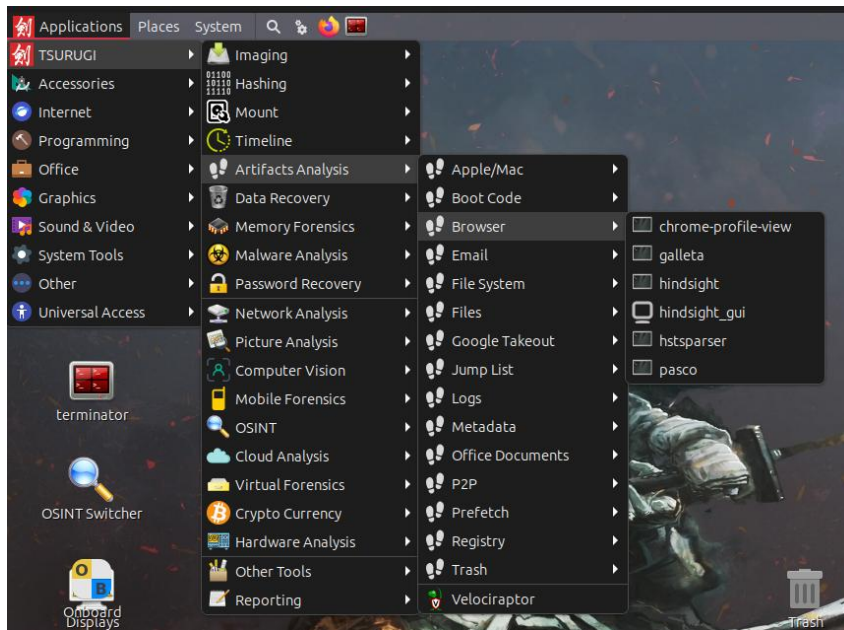
Environment Setup

(#1) First thing I wanted to confirm was what system I was on and how it was connected. Then I had to check that it had internet access, just getting all of the basics out of the way before I start.

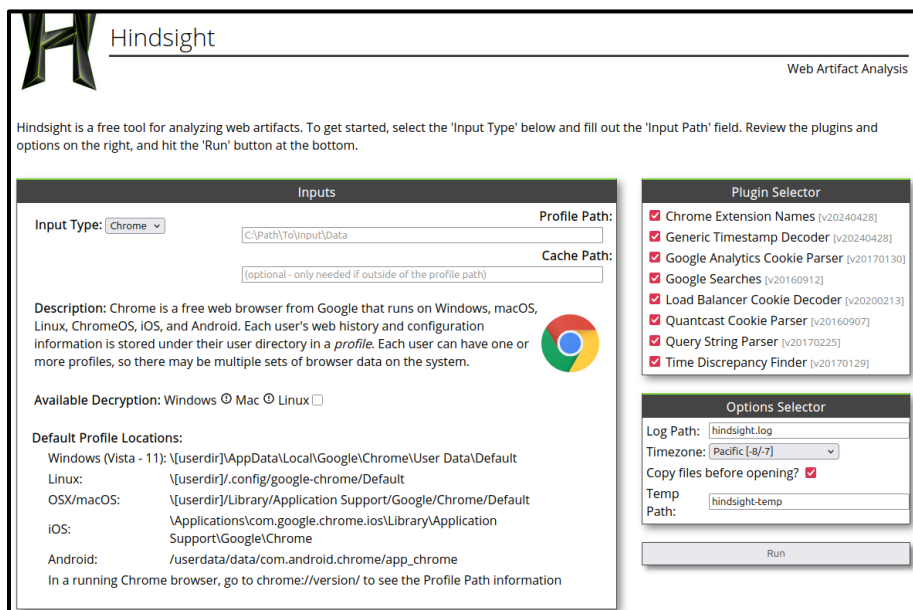
```
tsurugi@tsurugi:~$ whoami
tsurugi
tsurugi@tsurugi:~$ hostnamectl
  Static hostname: tsurugi
        Icon name: computer-vm
        Chassis: vm
        Machine ID: c27d4f84536f43d088d7cd51ddab30c4
        Boot ID: 8a2af1a0b2bd432ab366f6b5c8137e7b
  Virtualization: oracle
Operating System: Tsurugi Linux 26.03
        Kernel: Linux 6.19.10-tsurugi
  Architecture: x86-64
  Hardware Vendor: innotek GmbH
  Hardware Model: VirtualBox
Firmware Version: VirtualBox
  Firmware Date: Fri 2006-12-01
  Firmware Age: 19y 9month 3w 2d
tsurugi@tsurugi:~$
```

```
tsurugi@tsurugi: ~ 78x23
tsurugi@tsurugi:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s17: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:2f:79 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s17
        valid_lft 85043sec preferred_lft 85043sec
    inet6 fd17:625c:f037:2:c3f9:a8b8:c7c5:8cc2/64 scope global temporary dynamic
        valid_lft 86390sec preferred_lft 14390sec
    inet6 fd17:625c:f037:2:68dc:a393:d16a:7b37/64 scope global dynamic mngttmpa
        valid_lft 86390sec preferred_lft 14390sec
    inet6 fe80::8ee1:54a5:a92f:afcf/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
tsurugi@tsurugi:~$
```

(#2) Gosh, there are a lot of options here! I could have done this on my Kali setup, but this is why I was recommended Tsurugi in the first place. Look at all the DFIR applications!!



Since 'Hindsight' wanted us to use Chrome, and that wasn't preinstalled on the VM, I decided to instead go find a sample dataset on Github that I could use for practice. I found moain2026/ digital-forensics-lab with the help of GPT and downloaded the 'History_EVIDENCE' to load into Hindsight.



Troubleshooting

I ran into a bit of technical trouble that I had to troubleshoot. I'll throw that in here too, because mistakes are part of learning. (#3) So, while I was trying to access the evidence I

had just downloaded I created a directory. If you take a look at the photo for this one you may see what I failed to notice the first couple of tries. It's always the little things that trip us up. Well, I misspelled 'hindsight' as 'hindight' without the 'S' and this frustrated me to no end for about 30 mins..

```
tsurugi@tsurugi: ~/Downloads/hindsight-evidence 109x36

tsurugi@tsurugi:~$ cd ~/Downloads
tsurugi@tsurugi:~/Downloads$ mkdir hindight-evidence
tsurugi@tsurugi:~/Downloads$ mv History_EVIDENCE.db hindsight-evidence/History
mv: cannot move 'History_EVIDENCE.db' to 'hindsight-evidence/History': No such file or directory
tsurugi@tsurugi:~/Downloads$ ls -lh ~/Downloads
total 164K
drwxrwxr-x 2 tsurugi tsurugi 4.0K Sep 25 15:14 hindight-evidence
-rw-rw-r-- 1 tsurugi tsurugi 160K Sep 25 15:08 History_EVIDENCE.db
tsurugi@tsurugi:~/Downloads$ ls -lh ~/Downloads/hindsight-evidence
ls: cannot access '/home/tsurugi/Downloads/hindsight-evidence': No such file or directory
tsurugi@tsurugi:~/Downloads$ mv ~/Downloads/History_EVIDENCE.db ~/Downloads/hindight-evidence/History
tsurugi@tsurugi:~/Downloads$ ls -lb
total 164
drwxrwxr-x 2 tsurugi tsurugi 4096 Sep 25 15:14 hindight-evidence
-rw-rw-r-- 1 tsurugi tsurugi 163840 Sep 25 15:08 History_EVIDENCE.db
tsurugi@tsurugi:~/Downloads$ cd ~/Downloads
tsurugi@tsurugi:~/Downloads$ ls
hindight-evidence History_EVIDENCE.db
tsurugi@tsurugi:~/Downloads$ cd hindsight-evidence
bash: cd: hindsight-evidence: No such file or directory
tsurugi@tsurugi:~/Downloads$ rm hindight-evidence
rm: cannot remove 'hindight-evidence': Is a directory
tsurugi@tsurugi:~/Downloads$ rmdir hindight-evidence
rmdir: failed to remove 'hindight-evidence': No such file or directory
tsurugi@tsurugi:~/Downloads$ rmdir hindight-evidence
tsurugi@tsurugi:~/Downloads$ mkdir hindsight-evidence
tsurugi@tsurugi:~/Downloads$ ls
hindsight-evidence History_EVIDENCE.db
tsurugi@tsurugi:~/Downloads$ mv History_EVIDENCE.db hindsight-evidence
tsurugi@tsurugi:~/Downloads$ ls
hindsight-evidence
tsurugi@tsurugi:~/Downloads$ cd hindsight-evidence
tsurugi@tsurugi:~/Downloads/hindsight-evidence$ ls
History_EVIDENCE.db
tsurugi@tsurugi:~/Downloads/hindsight-evidence$
```

Investigation:

Ok, moving on! I now have a Chrome browser history (#4). What I want to answer is:

- What websites/services they visited
- What they searched for
- The approximate timeline/order of activity
- Anything that looks interesting or suspicious

History Forensics (v2025.03)		
Timestamp (US/Pacific)	URL	Title / Name / Status
2026-09-13 19:03:04.407	https://www.ksu.edu.sa/	جامعة الملك سعود King Saud Univ
2026-09-13 19:03:04.407	https://ksu.edu.sa/	جامعة الملك سعود King Saud Univ
2026-09-13 19:03:05.327	https://www.alriyadh.com/	جريدة الرياض
2026-09-13 19:03:08.934	https://www.sama.gov.sa/	البنك المركزي السعودي
2026-09-13 19:03:08.934	https://www.sama.gov.sa/ar-sa/Pages/default.aspx	البنك المركزي السعودي
2026-09-13 19:03:11.419	https://www.alrajhibank.com.sa/	الأفراد مصرف الراجحي
2026-09-13 19:03:14.218	https://mail.google.com/	Gmail
2026-09-13 19:03:14.218	https://mail.google.com/mail/	Gmail
2026-09-13 19:03:14.218	https://mail.google.com/mail/u/0/	Gmail
2026-09-13 19:03:14.218	https://accounts.google.com/ServiceLogin?service=mail&passive=120	Gmail
2026-09-13 19:03:14.218	https://accounts.google.com/v3/signin/identifier?continue=https:	Gmail
2026-09-13 19:03:16.761	https://www.aljazeera.net/	بزة نت: آخر أخبار اليوم حول العالم
2026-09-13 19:03:20.092	https://github.com/	GitHub · Change is constant. Git
2026-09-13 19:03:29.232	https://www.absher.sa/	أبشر-Absher
2026-09-13 19:03:29.232	https://www.absher.sa/portal/landing.html?v=1	أبشر-Absher
2026-09-13 19:03:31.016	https://drive.google.com/	Google Drive: Share Files Online
2026-09-13 19:03:31.016	https://accounts.google.com/ServiceLogin?service=wise&passive=120	Google Drive: Share Files Online
2026-09-13 19:03:31.016	https://www.google.com/intl/en-US/drive/	Google Drive: Share Files Online
2026-09-13 19:03:31.016	https://workspace.google.com/intl/en-US/products/drive/	Google Drive: Share Files Online

First im going to try to identify the websites that were visited.

- The first one is provided for me by Hindsight; King Saud University. This can be inferred because of the **.edu.sa** in the URL.
- **alriyadh.com** hmm.. the title for this one is in Arabic. I'm going to come back to this one.
- **sama.gov.sa** looks to be a Saudi government website.
- **alrajhibank.com.sa** appears to be a Saudi bank
- **mail.google.com** is an easy one, as is **accounts.google.com**. There are some interesting points in this one, but for now im just identifying web addresses.
- **Aljazeera.net** this is one I don't know and the title is in Arabic so we will have to do some digging.
- **Github.com** I may have heard of this site before :P
- **absher.sa** ok, another Saudi cite but again the title doesn't tell me much.
- Then I see google **drive**, google **accounts**, and google **workspace**.

Alrighty, that was the first glance. Now let's dig into the ones I didn't know. For this I basically just did a google search of each website to get a general idea.

- **Alriyadh** = News site
- **sama.gov.sa** = Saudi Central Bank
- **alrajhibank.com.sa** =an Islamic bank
- **Aljazeera.net** = Arabic news site
- **absher.sa** = Saudi Arabian Ministry of Interior

The timeline of events begins on 9/13/26 at 19:03:04 the user went to the website for King Saud University and moved very quickly. Literally, in under 30 seconds this entire series of events transpires.

They visit the University page, a new site, and 2 prominent banks back-to-back.

Then there is google mail, with 'mail.com/u/0/' referring to a login page and the 'v3/signin/identifier' being the google authentication framework.

In quick succession they then visit github, another news site, and a government site.

Just after that, the user opens a shared drive, and uses google Workspace to search for products. I went to this site myself just to see what would be available to view from the products page and it is mainly application integrations and productivity upgrades.

History Forensics (v2025.03)		
Timestamp (US/Pacific)	URL	Title / Name / Status
2026-09-13 19:03:04.407	https://www.ksu.edu.sa/	جامعة الملك سعود King Saud Univ
2026-09-13 19:03:04.407	https://ksu.edu.sa/	جامعة الملك سعود King Saud Univ
2026-09-13 19:03:05.327	https://www.alriyadh.com/	جريدة الرياض
2026-09-13 19:03:08.934	https://www.sama.gov.sa/	البنك المركزي السعودي
2026-09-13 19:03:08.934	https://www.sama.gov.sa/ar-sa/Pages/default.aspx	البنك المركزي السعودي
2026-09-13 19:03:11.419	https://www.alrajhibank.com.sa/	الأفراد مصرف الراجحي
2026-09-13 19:03:14.218	https://mail.google.com/	Gmail
2026-09-13 19:03:14.218	https://mail.google.com/mail/	Gmail
2026-09-13 19:03:14.218	https://mail.google.com/mail/u/0/	Gmail
2026-09-13 19:03:14.218	https://accounts.google.com/ServiceLogin?service=mail&passive=120	Gmail
2026-09-13 19:03:14.218	https://accounts.google.com/v3/signin/identifier?continue=https:	Gmail
2026-09-13 19:03:16.761	https://www.aljazeera.net/	بزة نت: آخر أخبار اليوم حول العالم
2026-09-13 19:03:20.092	https://github.com/	GitHub · Change is constant. Git
2026-09-13 19:03:29.232	https://www.absher.sa/	أبشر- أبشر
2026-09-13 19:03:29.232	https://www.absher.sa/portal/landing.html?v=1	أبشر- أبشر
2026-09-13 19:03:31.016	https://drive.google.com/	Google Drive: Share Files Online
2026-09-13 19:03:31.016	https://accounts.google.com/ServiceLogin?service=wise&passive=120	Google Drive: Share Files Online
2026-09-13 19:03:31.016	https://www.google.com/intl/en-US/drive/	Google Drive: Share Files Online
2026-09-13 19:03:31.016	https://workspace.google.com/intl/en-US/products/drive/	Google Drive: Share Files Online

Findings & Conclusion

Alrighty, so what did I actually learn from all of this?

Hindsight was able to pull 19 URLs from the Chrome history I gave it, and from there I was able to get a pretty good idea of what the user was doing. They visited several Saudi websites, including a university, two banks, and two news sites, and government services. They also accessed Gmail, GitHub, and Google Drive/Workspace.

What stood out to me the most was how fast all of this happened. The entire history I was given takes place in less than 30 seconds. Ultimately, there really isn't enough information here to say *why* they were visiting these sites or that they were doing anything suspicious.

If I wanted to keep investigating, I would definitely want more evidence. A longer browser history, searches, downloads, cookies, or other system activity would give me a lot more context about what the user was actually doing.

Overall, I think my biggest takeaway from this lab is that finding the evidence is only half the job. I also have to be careful about what I can actually prove from it versus what I *think* might have happened.